of 1–10 % for the A-values, meaning the lowest increase in posterior a theory can gain from a piece of evidence is 1%, and the highest gain is a 10% increase. Given that the A-Score "Accepted" is the natural max, that means that a piece of evidence with the A-Score "Accepted" should increase the posterior by 10 % (the A-value is 10%). Similarly, a piece of evidence with the lowest ordinal A-Score ("Coherent but untestable") should increase the posterior by 1 % (the A-value is 1%). In this way the highest and lowest ordinals anchor any scale we decide on. But how do we non-arbitrarily set the A-value of the middle ordinal(s)? Disagreement seems possible on this question. For instance, one might suggest that the A-value of the middle ordinal should be in the middle (or close to) between the min and max, say 5 % on the scale used here. Others might disagree and argue that the testability difference between "Coherent and testable" and "Coherent but untestable" is of such significance that the A-value of the middle ordinal should be closer to 8 or 9 percent, rather than in the middle. So, how do we determine what the A-value of the middle ordinal (e.g., evidence scored as "Coherent and testable") should be? Critically, we need to deal with *arbitrariness* and not bias the comparison against any theory. This is especially important because the A-value of the middle ordinal influences the posteriors (because it determines the increase a theory gains from a piece of evidence with the A-Score "Coherent and testable." see Figure 3). In QBE this issue is solved by refusing to fix the middle ordinal to one value. Instead the idea is to calculate the entire dataset for each possible value of the middle ordinal (e.g., using natural numbers 2 through 9 in our example data) and let the collective set of posteriors form the basis for our comparison of theories. The same solution is applied to the R-score. The example data here uses R-values of one to ten,[22] and calculates the dataset with each possible R-value for the middle ordinal ("Medium Replication"). Consequently, in our example here, the output of the quantification for a given theory is a set 64 posteriors (8*8) reflecting each combination of the possible middle ordinals of the A-value (2–9%) and the R-value (2–9).

But what about a case where someone wants to deploy more than three ordinals? In these cases there will be two (or more) middle ordinals rather than one. While this increases the combinations in terms of the number of posteriors that need to be calculated, there is nothing inherently problematic with this. Naturally, because two or more middle ordinals are ranked *qua* ordinals, they constrain each other in terms of the values each can have. To illustrate, using a 1–10 scale, if the lower of two middle ordinals has a value of 4, this constrains the possible values of the higher middle ordinal to the numbers [5,6,7,8,9]. In sum, the methodology can easily accommodate cases where more than one middle ordinal is deployed. While the number of posteriors that will be calculated for a theory will increase with the number of middle ordinals, this increase is trivial, and not such that it poses a problem for the methodology.

4.5 Quantification and comparison

The objective in this section is to demonstrate the proposed methodology. Initially, as just discussed the exact scaling of the parameters is inconsequential for the comparison as long as it is applied to every theory being compared.[23] Unfortunately, there is an immediate obstacle to the demonstration in that there are no datasets on which to demonstrate the methodology. We simply do not have a full view of all — and which — phenomena are claimed in favor of any theory (let alone all theories). While some work has been done on this (Kirkeby-Hinrup and Fazekas, 2021), there is considerable way to go before we have complete compilations for every theory, and have separately assessed each proposed piece of empirical support to derive an A-score and R-score (and any other score we may think of, see below). To address this issue, I wrote a small piece of software[24] that generates random datasets for hypothetical theories. The datasets were set to contain between 20 and 40 phenomena, that each had an A-score and an R-score. Generation of A-and R-scores was weighted to output comparatively fewer of the highest ordinal (see footnote 20) to account for the fact that there currently is not much knock down evidence. I generated four hypothetical theories [A,B,C,D], whose resulting datasets sorted in 3 (A-score) by 3 (R-score) matrices are shown in Table 2. I then ran the updating mechanism (using a scaling that allowed the posteriors to stay within Bayesian 0 and 1 bounds) on the datasets of the four hypothetical theories.

For determining the likelihood, a scale for the A-value of one to 10 % was used, meaning the highest A-score ordinal ("Accepted") implied a 10 % increase, and the lowest ordinal ("Coherent but untestable") a 1 % increase, with the middle ordinal ("Coherent and testable") occupying each of the intermediary steps (2–9 percent). The marginal also used a one to ten scale for the R-value. However, the one to ten scale of the R-value was not percent, but rather hundreds. Again, the highest R-score ordinal ("High") being 10/100, the lowest ("Low") being 1/100, and the middle ordinal ("Medium") occupying the intermediate steps (2–9/100). To avoid the counterintuitive result that well replicated studies might decrease a posterior, the marginal was calculated as one minus the R-value (Figure 4). The initial prior was set at 0.04 (based on the assumption that there are at least 25 viable competing theories). The updating itself consisted in iterating through the list of proposed evidence (phenomena) deriving a new posterior after the inclusion of each phenomenon, and that posterior becoming the new prior when updating with the next phenomenon. The end result, with the entire set of proposed evidence processed, is a posterior *given all the evidence*.

Now, because of the way R-values and A-values of the middle ordinals are modeled in QBE the updating has to be carried out with each of possible combination of R-value and A-value. In this example, the result after updating is a dataset for each theory consisting in sixty-four posteriors; *viz* one for each tested combination of values of the middle ordinal of the A-score (2–9%) and R-score (2-9/100). When ordering the datasets according to size of the posterior and plotting them on a graph, (Figure 3) the impact of the value of the middle ordinals of the A-score and R-score is evident. Similarly, by calculating the mean and standard deviation of all the posteriors (Table 3) we can represent the probability that a theory has a given posterior (Figure 5).

22 Again: arbitrarily chosen since the exact scaling is inconsequential for the comparison. The important thing is to keep it the same for all theories.

23 In the previous sections, I have mentioned at the parameters that will be deployed here, but I will reiterate them the first time they appear.

24 Not on github, but open source in the sense that I will give you the code if you send me an email.

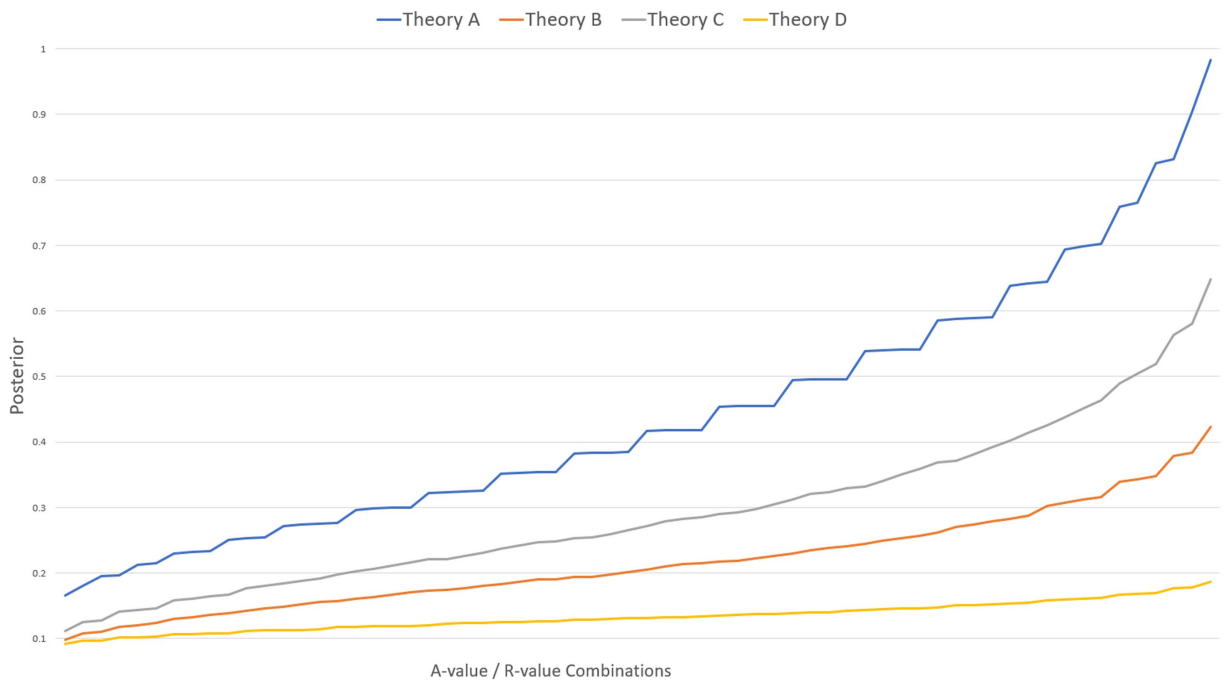

FIGURE 3
Impact of middle ordinal combinations (x) on posterior (y).

TABLE 2 Hypothetical datasets of theories A, B, C, and D.

		Accepted	Coherent and testable	Coherent but untestable
	Theory B		**A-score**	
R-score	High	1	0	4
	Medium	1	7	8
	Low	0	2	3
	Theory B		**A-score**	
R-score	High	0	2	1
	Medium	0	6	3
	Low	0	4	4
	Theory C		**A-score**	
R-score	High	0	2	0
	Medium	0	5	8
	Low	1	5	4
	Theory D		**A-score**	
R-score	High	1	1	1
	Medium	1	1	3
	Low	0	3	3

While helpful — at a glance — to get an impression of where each theory stands, these are merely ways of depicting the data, and do not suffice as an actual *comparison*.

There are several different ways of going about comparing the numbers. I will next consider a few options. The most straightforward approach would be to compare directly the mean posteriors of the theories, i.e., collapse the set of posteriors from each theory into a mean posterior for each theory and compare them. The mean posteriors — in themselves — allow for straightforward comparison of the theories on the bases of their respective posteriors (Table 3).

$$\text{POSTERIOR} \quad P(H|e_1 \ldots e_n) = \frac{\text{LIKELIHOOD} \quad \text{PRIOR} \quad \text{A-value} \; P(H)}{1 - \text{R-value} \quad \text{MARGINAL}}$$

FIGURE 4
Example updating mechanism.

TABLE 3 Means and standard deviation.

	A	B	C	D
Mean posterior*	0.4409	0.2157	0.292	0.1329
Standard deviation*	0.1940	0.0744	0.1218	0.0219

*Rounded to 4 decimals.

A similar second option could be to take the graphs in Figure 3 and compare the areas under the curve (this could be refined using smaller increments for the calculated A-and R-values and by deploying integrals). However, a more interesting third option may be Z-score comparison (Figure 6). The idea behind Z-scores is to use the mean and SD of posteriors of all theories to create an anchor for how much support a given theory has as compared to

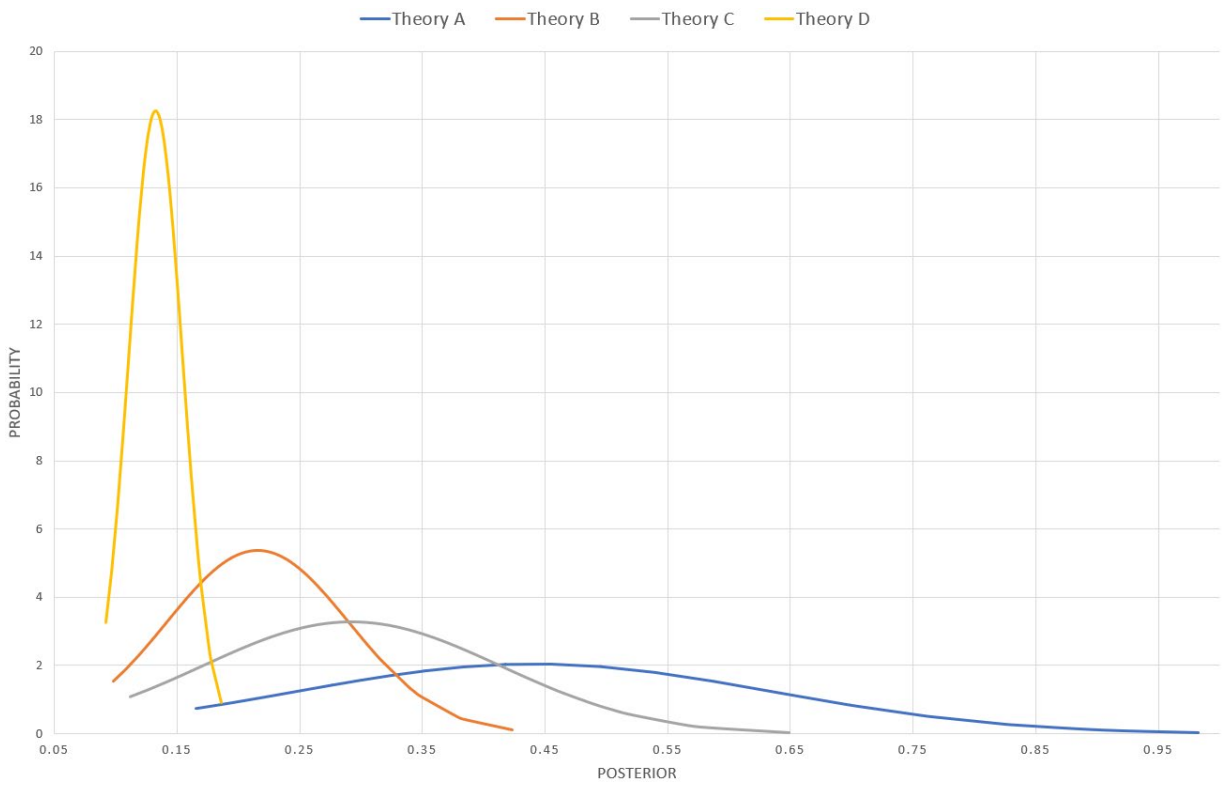

FIGURE 5
Normal distribution of probability (y-axis) that a theory has a given posterior (x-axis).

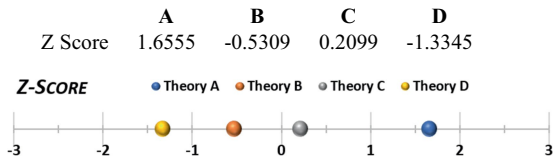

FIGURE 6
Z-score comparison.

the average support theories have. The Z-score shows how many standard deviations a theory's support is from the average. Ranging from-3 to +3, positive Z-scores indicate good support relative to the norm. For comparisons of two concrete theories, a fourth option could be deploying t-tests, either one-tailed (pairwise comparison), or two-tailed, to assess whether the empirical support for theories across the field is truly different. Finally, one could compare theories directly against each other using pairwise ratios of the mean posteriors (Table 4).

One important feature of each of these comparison options is that they are all independent of our scaling choices in the sense that — while it is nice that our data fits within the Bayesian bounds — the comparisons themselves do not depend on this (i.e., we could still do mean posteriors, Z-scores, ratio comparisons etc. if the posteriors were higher than 1). This gives significant flexibility to our choices with respect to scaling, and counters potential issues with arbitrariness in this regard.

4.6 Arbitrariness in scoring, ordinals, and the updating mechanism

In the above, *arbitrariness* has been prevented at every turn, yet three issues remain in this regard that need to be addressed. The first (and most critical) of these is *who* gets to determine the A-score of a piece of proposed empirical support. Given that the A-score directly impacts the amount of support gained from a piece of evidence, if this is left at the whim of the comparer, the whole process is undermined. The solution is straightforward: in the scoring it is necessary to engage with the original authors of a given piece of proposed empirical support.[25] Such engagement serves to make certain that the A-score assigned to each piece of empirical evidence is corroborated by the views of the original authors.[26] Furthermore, the engagement with the proponents of a given piece of empirical support affords them opportunity to clear up misunderstandings, make corrections, or further specify their argument in light of problems exposed (that result in an A-score they disagree with), or questions that arose in the case-by-case analysis. It is also important to recognize that novel experimental paradigms may impact A-scores. One recent example of this pertains to the pneumatic drill example given in favor of the

25 As well as other relevant proponents of the theory.

26 Remember, their conceptual framework has to be taken for granted, which makes them the authorities on interpretation of their proposed evidence, as noted in section three.